

Sharp lower-end simplex stability from integrated witnesses and projection caps

AI-assisted research supplement to entry005

7 October 2026

Result. For every fixed $d \geq 3$ and every maximum-volume inscribed simplex S ,

$$E(K, S) \leq G_d^\# \left(a(K) - \frac{1}{d+1} \right)^{1/(d-1)}.$$

All constants below are explicit. The exponent matches the exact truncation obstruction. The additional geometric step uses centering to compare normalized projection volumes, controls scale by first variation, and then puts a cap in a $(d-1)$ -dimensional projection. No inverse-Minkowski stability theorem is an input. The supplied explicit $1/d$ theorem is preserved as a separate source, with its constants unchanged. An independent analytic model audit passed, with independently implemented finite checks. No human peer review or proof-assistant formalization is claimed. No priority or best-known-result claim is made.

1 Definitions and explicit constants

For a convex body $K \subset \mathbb{R}^d$, let ΠK denote the projection body with $h_{\Pi K}(u) = |K \mid u^\perp|_{d-1}$ for unit u , and put

$$R(K) = \frac{|\Pi K|}{|K|^{d-1}}, \quad a(K) = \left(\frac{d}{d+1} \right)^d \frac{R(\mathcal{P}K)}{R(K)} - 1, \quad e(K) = a(K) - \frac{1}{d+1}.$$

Here $\mathcal{P}K$ is a pyramid over K , as in entry005. For a simplex $S \subset K$ with centroid z_S , define

$$E(K, S) = \inf\{t \geq 0 : K \subset z_S + (1+t)(S - z_S)\}.$$

For the constants set

$$R_0 = d(d+1), \quad b = \frac{1}{4(dR_0)^d}, \quad M = b^{-1}, \quad (1)$$

$$Q = (d+1)(d+2)8^d b^{-4d}, \quad m = d-1, \quad L = m(M+1), \quad (2)$$

$$J = \frac{d}{2}(2R_0)^d[1 + M + d2^{d-1}M], \quad (3)$$

$$r_\# = \min \left\{ b, \frac{1}{J(8MdL)^m} \right\}, \quad e_\# = \frac{r_\#}{Q(d+1)}, \quad (4)$$

$$A_d^\# = 4MdL[JQ(d+1)]^{1/m}, \quad G_d^\# = \max\{A_d^\#, (R_0-1)e_\#^{-1/m}\}. \quad (5)$$

All constants other than the displayed m th roots are positive rationals. They are extremely poor numerically and depend on dimension.

Theorem 1. For every convex body $K \subset \mathbb{R}^d$, $d \geq 3$, and every maximum-volume inscribed simplex S ,

$$E(K, S) \leq G_d^\sharp e(K)^{1/(d-1)}.$$

If $0 \leq e(K) \leq e_\sharp$, the constant $A_d^\sharp$ applies, and $A_d^\sharp e_\sharp^{1/(d-1)} \leq 1/2$. No larger power can hold uniformly near $e = 0$, even if one permits a dimension-dependent constant.

2 Normalization and the cone law

Fix an arbitrary maximum simplex and map it affinely to a regular simplex Δ with centroid zero and inradius one. Its vertices have norm d . Replacing a vertex by $x \in K$ multiplies the simplex volume by the absolute value of the corresponding barycentric coordinate. Maximality therefore gives $|\alpha_i(x)| \leq 1$, and

$$B_2^d \subset \Delta \subset K \subset R_0 B_2^d. \quad (6)$$

Both a and E are preserved by this affine normalization.

Write h_K for the support function and S_K for surface area measure. Push the probability measure $h_K(u) dS_K(u)/(d|K|)$ forward by $u \mapsto u/h_K(u)$; call the resulting law ν . The classical identities $\int u dS_K(u) = 0$ and $\int h_K dS_K = d|K|$ show that ν is centered. It is supported on $\partial K^\circ \subset B_2^d$. For independent samples from ν , define

$$A = \mathbb{E}|\det(X_1, \dots, X_d)|, \quad B = \mathbb{E}|\det((X_1, 1), \dots, (X_{d+1}, 1))|.$$

The entry005 cone-law representation is

$$a(K) = \frac{B}{(d+1)A}, \quad D := B - A = (d+1)Ae(K) \leq (d+1)e(K). \quad (7)$$

Here $0 < A \leq 1$, the upper bound following from Hadamard's inequality. For completeness, the representation for polytopes follows by expanding the two determinant expectations over distinct facets, with the $d!$ and $(d+1)!$ permutation factors. Polytope approximation, uniform support-function convergence and weak convergence of surface area measures give the arbitrary-body identity; all laws stay in a common compact set. This representation, already established in entry005 v3, is the only entry005-specific identity imported here.

For a body H containing zero in its interior, its cone law satisfies the Cauchy identity

$$\int |u \cdot x| d\nu_H(x) = \frac{2|H| |u^\perp|_{d-1}}{d|H|} \quad (|u| = 1). \quad (8)$$

Indeed the factors h_H cancel in the pushforward integral, leaving $\int |u \cdot n| dS_H(n)/(d|H|)$. Centering, Cauchy's formula, and (6) give

$$\mathbb{E}(-u \cdot X)_+ = \frac{|K| |u^\perp|_{d-1}}{d|K|} \geq \frac{(2/d)^{d-1}}{d(2R_0)^d} = 2b. \quad (9)$$

The projected unit ball contains a cube of side $2/d$.

3 The integrated assignment bound, including nonatomic laws

This section reproduces the probabilistic argument in the supplied readable fallback so that the quantitative proof is complete. For an ordered base tuple $\mathbf{x} = (x_1, \dots, x_d)$ put

$$F_{\mathbf{x}}(y) = \det((x_1, 1), \dots, (x_d, 1), (y, 1)), \quad P(\mathbf{x}) = \mathbb{E}(F_{\mathbf{x}}(X))_+, \quad N(\mathbf{x}) = \mathbb{E}(-F_{\mathbf{x}}(X))_+.$$

Since $\mathbb{E}X = 0$, $\mathbb{E}F_{\mathbf{x}}(X)$ is the horizontal determinant, up to sign. Subtracting its absolute value from $\mathbb{E}|F_{\mathbf{x}}(X)|$ and integrating gives

$$D = 2\mathbb{E}_{\mathbf{X}} \min\{P(\mathbf{X}), N(\mathbf{X})\} \geq 0. \quad (10)$$

No nonsingularity assumption on a sampled base is needed.

Use the continuous nonnegative witness

$$\psi_{\mathbf{x}}(y, z) = \min\{(F_{\mathbf{x}}(y))_+, (-F_{\mathbf{x}}(z))_+\} + \min\{(-F_{\mathbf{x}}(y))_+, (F_{\mathbf{x}}(z))_+\}.$$

For independent Y, Z each of its minima has expectation at most $\min(P, N)$; hence

$$\mathbb{E}_{\mathbf{X}, Y, Z} \psi_{\mathbf{X}}(Y, Z) \leq D. \quad (11)$$

Lemma 2 (Integrated assignment). *For any centered law on B_2^d satisfying (9), there are affinely independent $w_0, \dots, w_d \in \text{supp } \nu$ and a measurable assignment $X \mapsto w_{I(X)}$ with*

$$h := \mathbb{E}\|X - w_{I(X)}\| \leq QD.$$

If $QD \leq b$, then $bB_2^d \subset T := \text{conv}(w_0, \dots, w_d) \subset B_2^d$.

Proof. Let $W_0, \dots, W_d$ be independent samples and let $V(W) = |\det((W_0, 1), \dots, (W_d, 1))|$. For $\Sigma = \mathbb{E}XX^\top$, centering and (9) give $\mathbb{E}|u \cdot X| \geq 4b$ and $u^\top \Sigma u \geq 16b^2$. Expanding determinants and using independence yields

$$\mathbb{E}V^2 = (d+1)! \det \Sigma \geq (d+1)! 16^d b^{2d} \geq 2b^{2d}.$$

The lifted second-moment matrix is $\text{diag}(\Sigma, 1)$; the determinant expansion gives $(d+1)!$ times its determinant. Subtracting one lifted column from the others and applying Hadamard gives $V \leq 2^d$. Consequently, with

$$v_0 = b^d, \quad q_0 = b^{2d}/4^d, \quad \mathcal{E} = \{W : V(W) \geq v_0\},$$

we have $\mathbb{P}(\mathcal{E}) \geq q_0$, since $\mathbb{E}V^2 \leq b^{2d} + 4^d \mathbb{P}(\mathcal{E})$.

For each i use the other d anchors as base and test w_i, x . For each pair $i < j$, use x and the other $d-1$ anchors as base and test w_i, w_j . Let $H(W)$ be the integral in x of their sum; there are $N_d = (d+1)(d+2)/2$ summands. Each uses $d+2$ independent samples before conditioning, so (11) gives $\mathbb{E}H(W) \leq N_d D$. Conditioning on $\mathcal{E}$, choose anchors with $H(W) \leq N_d D/q_0$. This selection is valid for nonatomic laws: H is continuous by dominated convergence, $\mathcal{E} \cap (\text{supp } \nu)^{d+1}$ is compact with positive probability, and its minimum is at most its conditional average.

Let $\alpha_i(x)$ be barycentric coordinates in the selected simplex. The first witness is

$$V \min\{1, (-\alpha_i(x))_+\}.$$

The pair witness is at least $V \min\{(\alpha_i(x))_+, (\alpha_j(x))_+\}$: when both coefficients are positive the tested determinants have opposite signs and magnitudes $V\alpha_j(x)$ and $V\alpha_i(x)$. Thus

$$\Phi := \mathbb{E} \left[\sum_i \min\{1, (-\alpha_i(X))_+\} + \sum_{i < j} \min\{(\alpha_i(X))_+, (\alpha_j(X))_+\} \right] \leq \frac{N_d D}{q_0 v_0}.$$

Replacement determinants are at most 2^d , so $|\alpha_i(x)| \leq L_0 := 2^d/v_0$ for $x \in B_2^d$. Assign x to its largest positive coefficient, with deterministic tie-breaking, and call this index r . Writing $N_x = \sum_i (-\alpha_i(x))_+$ and $U_x = \sum_{i \neq r} (\alpha_i(x))_+$, we have

$$N_x \leq L_0 \sum_i \min\{1, (-\alpha_i(x))_+\}, \quad U_x \leq \sum_{i < j} \min\{(\alpha_i(x))_+, (\alpha_j(x))_+\}.$$

For the second inequality use the pairs containing r . The diameter bound gives $\|x - w_r\| \leq 2 \sum_{i \neq r} |\alpha_i(x)| = 2(N_x + U_x)$. Therefore

$$h \leq 2L_0 \Phi \leq \frac{2L_0 N_d}{q_0 v_0} D = (d+1)(d+2)8^d b^{-4d} D = QD.$$

Finally, $(u \cdot x)_+$ is one-Lipschitz and its expectation is at least $2b$. The assigned law has expectation at least $2b - h \geq b$ if $QD \leq b$. In every direction an anchor therefore has $u \cdot w_i \geq b$, proving $bB_2^d \subset T$. The upper inclusion is automatic. $\square$

4 Centering controls normalized projection volumes

Assume now $QD \leq b$, select the anchors from Lemma 2, and put $P = T^\circ$. It is an enclosing simplex:

$$B_2^d \subset K \subset P \subset MB_2^d. \quad (12)$$

Each $w_i \in \partial K^\circ$, so $h_K(w_i) = h_P(w_i) = 1$. Let q_i be the vertex of P opposite its facet with normal w_i , and let λ_i be the barycentric coordinates of zero in T . They are positive and sum to one. The cone law of P is

$$\nu_P = \sum_{i=0}^d \lambda_i \delta_{w_i}. \quad (13)$$

To see this, the polar images of the facet normals of P are precisely w_i . Its cone law is centered, so its probability weights are the unique barycentric coordinates of zero.

Write $p_i = \mathbb{P}(I(X) = i)$ and $c = \sum_i p_i w_i$. Centering of ν gives $\|c\| \leq h$. The barycentric coordinate functions of T have the exact formula

$$\alpha_i(x) = \lambda_i(1 - q_i \cdot x). \quad (14)$$

For $j \neq i$, $q_i \cdot w_j = 1$; taking the scalar product of $\sum_j \lambda_j w_j = 0$ with q_i gives $q_i \cdot w_i = 1 - 1/\lambda_i$. These values prove (14) on an affine basis. Consequently

$$p_i - \lambda_i = -\lambda_i q_i \cdot c, \quad \sum_i |p_i - \lambda_i| \leq M\|c\| \leq Mh. \quad (15)$$

For any unit u , the function $x \mapsto |u \cdot x|$ is one-Lipschitz and bounded by one on T . Comparing first with the assigned law and then with (13) gives

$$\left| \int |u \cdot x| d\nu(x) - \int |u \cdot x| d\nu_P(x) \right| \leq (1 + M)h.$$

By (8), if $\pi_H(u) = |H| u^\perp|_{d-1}$, then

$$\left| \frac{\pi_K(u)}{|K|} - \frac{\pi_P(u)}{|P|} \right| \leq \frac{d}{2}(1+M)h. \quad (16)$$

This is the additional information retained by the integrated assignment.

5 Scale control and absolute projection deficits

Since h_P is M -Lipschitz and equals one at every anchor,

$$0 \leq z := \int (h_P(x) - 1) d\nu(x) \leq Mh \leq 1.$$

The integral is $V(K[d-1], P)/|K|-1$. The volume first-variation formula and the Brunn–Minkowski inequality give Minkowski's first inequality, and hence

$$\frac{|P|}{|K|} \leq (1+z)^d, \quad |P| - |K| \leq d2^{d-1}Mh|K|. \quad (17)$$

The second bound uses $(1+z)^d - 1 \leq d2^{d-1}z$ for $0 \leq z \leq 1$. Explicitly, differentiating $|K+tP|^{1/d} \geq |K|^{1/d} + t|P|^{1/d}$ at $t=0+$ proves the mixed-volume inequality just used.

Since ν_P is supported on B_2^d , (8) gives $\pi_P(u)/|P| \leq d/2$. Using (16) and (17),

$$\begin{aligned} 0 \leq \pi_P(u) - \pi_K(u) &= \frac{\pi_P(u)}{|P|}(|P| - |K|) + |K| \left(\frac{\pi_P(u)}{|P|} - \frac{\pi_K(u)}{|K|} \right) \\ &\leq \frac{d}{2}|K|[d2^{d-1}M + 1 + M]h \leq Jh. \end{aligned}$$

The last step uses $|K| \leq (2R_0)^d$. We have thus obtained the uniform absolute projection bound

$$\sup_{|u|=1} \{\pi_P(u) - \pi_K(u)\} \leq Jh. \quad (18)$$

The nonnegativity uses the actual inclusion $K \subset P$.

6 A cap in one projection

Lemma 3 (Projection cap). *If $B_2^d \subset K \subset P \subset MB_2^d$, $d \geq 2$, and $\pi_P(u) - \pi_K(u) \leq \varepsilon$ in every direction, then*

$$d_H(K, P) \leq (d-1)(M+1)\varepsilon^{1/(d-1)}.$$

No simplex hypothesis is needed for this lemma.

Proof. Put $s = d_H(K, P)$. If $s = 0$ there is nothing to prove. Otherwise choose a farthest $q \in P$ and its metric projection $k \in K$, and set $n = (q - k)/s$. The projection optimality condition gives $n \cdot k = h_K(n)$, so $n \cdot q - h_K(n) = s$. Also $s \leq M$, since $0 \in K$. Choose a unit $u \perp n$ and project orthogonally onto $u^\perp$. Let $\bar{P}, \bar{K}, \bar{q}$ be the projected objects. The space has dimension $m = d-1$, n lies in it, $\bar{P}$ contains its unit ball and is contained in its ball of radius M , and $n \cdot \bar{q} - h_{\bar{K}}(n) = s$.

With $\tau = s/[2(M+1)] < 1$, convexity places the ball $(1-\tau)\bar{q} + \tau B_2^m$ inside $\bar{P}$. Its lowest n -coordinate is at least

$$n \cdot q - \tau(M+1) = h_{\bar{K}}(n) + s/2,$$

so it is disjoint from $\overline{K}$. A cube of side $2\tau/m$ fits in that ball. Therefore

$$\varepsilon \geq |\overline{P}| - |\overline{K}| \geq (2\tau/m)^m = \left[\frac{s}{(d-1)(M+1)} \right]^{d-1}.$$

This proves the claim. $\square$

Equations (18) and Lemma 3 yield the simplex-specific surface/assignment conversion

$$s = d_H(K, P) \leq L(Jh)^{1/m} \leq L[JQ(d+1)e(K)]^{1/m}. \quad (19)$$

It is stronger in exponent than the full-dimensional volume-cap conversion.

7 Retaining every initially chosen maximum simplex

If $0 \leq e \leq e_\sharp$, then $QD \leq Q(d+1)e \leq r_\sharp \leq b$ and (19) gives $s \leq 1/(8Md)$. Because $h_P \geq 1$ and $h_K \geq h_P - s$, $(1-s)P \subset K$. The initially selected Δ is maximum, so

$$\frac{|\Delta|}{|P|} \geq (1-s)^d \geq 1 - ds.$$

Put $\delta = ds \leq 1/8$. The barycentric coordinates of the vertices of Δ in P form a nonnegative column-stochastic $(d+1) \times (d+1)$ matrix W with $|\det W| = |\Delta|/|P| \geq 1 - \delta$. Each column has norm at most one. Hadamard's inequality therefore gives squared norm at least $(1-\delta)^2$ in each column, so each column has an entry at least $1 - 2\delta$.

These dominant rows are distinct. Indeed, the permanent is the probability that independent row selections from the columns are all distinct and dominates $|\det W|$. A common dominant row in two columns would give

$$|\det W| \leq \text{per } W \leq 1 - (1 - 2\delta)^2 \leq 4\delta < 1 - \delta,$$

a contradiction. Match the vertices by these rows. Each vertex of Δ is within $2\delta \text{ diam } P \leq 4Md s$ of its matched vertex of P . Taking convex combinations and using $B_2^d \subset \Delta$ gives

$$K \subset P \subset \Delta + 4Md s B_2^d \subset (1 + 4Md s)\Delta.$$

By (19), $E(K, \Delta) \leq A_d^\sharp e^{1/m}$. This also covers $e = 0$: then $h = 0$, the projection deficits vanish, $s = 0$, and the same matrix argument gives zero excess. At the threshold,

$$A_d^\sharp e_\sharp^{1/m} = 4MdL(Jr_\sharp)^{1/m} \leq \frac{1}{2}.$$

For $e > e_\sharp$, (6) gives $K \subset R_0\Delta$ and hence $E(K, \Delta) \leq R_0 - 1$, which is covered by $G_d^\sharp$ in (5). Undoing the affine normalization restores the centroid of the arbitrary initial S . Thus the estimate holds for every maximum simplex.

8 Sharpness and the obstruction to using volume alone

The pinned truncation source gives the following exact calculation; the needed parts are derived here. In unnormalized coordinates let

$$P_0 = \text{conv}(0, e_1, \dots, e_d), \quad K_t = \{x_i \geq 0, t \leq \sum_i x_i \leq 1\}, \quad 0 < t < 1.$$

Put $q = t^{d-1}$ and $c = 1/(d-1)!$. Its facet area vectors and area times support numbers are

$$(u_i, b_i) = (-c(1-q)e_i, 0), \quad (u_+, b_+) = (c\mathbf{1}, c), \quad (u_-, b_-) = (-cq\mathbf{1}, -ct^d).$$

The polytope determinant formula for entry005 is $a = L_{\det}/(d|K_t|H_{\det})$. Horizontal d -minors and lifted $(d+1)$ -minors give respectively

$$\begin{aligned} H_{\det} &= c^d(1-q)^{d-1}[d+1+(d-1)q], \\ L_{\det} &= c^{d+1}(1-q)^{d-1}[1+(d-1)q-(d-1)t^d-qt^d]. \end{aligned}$$

For the first sum the all-coordinate subset contributes $c^d(1-q)^d$, while each subset with one of the parallel top/bottom normals contributes $c^d(1-q)^{d-1}$ or $c^d q(1-q)^{d-1}$. For the second sum, the all-coordinate subsets with top or bottom contribute $c^{d+1}(1-q)^d$ and $c^{d+1}(1-q)^d t^d$; each of the d remaining subsets contributes $c^{d+1}(1-q)^{d-1}(q-t^d)$. Since $|K_t| = (1-t^d)/d!$, subtraction of $1/(d+1)$ gives

$$e(K_t) = \frac{t^{d-1}[d(d-1)-(d+1)(d-2)t-2t^d]}{(d+1)(1-t^d)[d+1+(d-1)t^{d-1}]}, \quad \lim_{t \downarrow 0} \frac{e(K_t)}{t^{d-1}} = \frac{d(d-1)}{(d+1)^2}. \quad (20)$$

Using nonpositive support numbers here is legitimate: this is a facet determinant calculation, not a probability law. Translation changes the lifted determinants by a row operation and leaves them unchanged.

The simplex $S_t = \text{conv}(e_1, \dots, e_d, te_1)$ is maximum. Indeed a nondegenerate vertex simplex must represent all d coordinate rays, with one ray doubled. If k of its other selected vertices are bottom vertices, its lifted determinant has magnitude $(1-t)^k$. The maximum is $1-t$. For arbitrary inscribed vertices, separate affine linearity and the triangle inequality bound their determinant by that same vertex maximum. Thus $|S_t| = (1-t)/d!$ is the global maximum.

Writing $s_x = \sum_i x_i$, barycentric coordinates in S_t are

$$\beta_0 = \frac{1-s_x}{1-t}, \quad \beta_1 = x_1 - t\beta_0, \quad \beta_i = x_i \quad (2 \leq i \leq d).$$

The minimum of β_1 on K_t is $-t$, attained at te_2 ; the other coordinates have minimum zero. Centroid dilation by factor $1+\epsilon$ is equivalent to $\beta_i \geq -\epsilon/(d+1)$. Hence

$$E(K_t, S_t) = (d+1)t.$$

Together with (20), this forbids every power $\alpha > 1/(d-1)$ in an every-maximum-simplex estimate. At the endpoint any such universal constant must satisfy

$$G_d \geq (d+1) \left[\frac{(d+1)^2}{d(d-1)} \right]^{1/(d-1)}.$$

There is also an exact obstruction to strengthening only the old scalar mixed-volume step. For this nested pair,

$$d_H(K_t, P_0) = \frac{t}{\sqrt{d}}, \quad \frac{V(K_t[d-1], P_0)}{|K_t|} - 1 = \frac{t^d}{1-t^d}.$$

The distance maximum occurs at the missing vertex zero; its nearest point is $(t/d)\mathbf{1}$. In the first-variation integral only the bottom facet has a support gap, namely $t/\sqrt{d}$, and its area is $\sqrt{d}t^{d-1}/(d-1)!$. This proves the mixed-volume formula. Thus an estimate using this scalar alone

cannot have a power greater than $1/d$. For all sufficiently small t , a fixed translation and dilation put these pairs in a position with a common inner ball and a fixed outer ball; the obstruction is therefore not caused by the coordinate origin. In contrast, for $u = (e_1 - e_2)/\sqrt{2}$, Cauchy’s formula gives

$$\pi_{P_0}(u) - \pi_{K_t}(u) = \frac{t^{d-1}}{\sqrt{2}(d-1)!}.$$

The top and bottom area vectors are perpendicular to u , and only the two coordinate facets contribute. This identifies exactly the information that the projection argument preserves.

9 Provenance and verification scope

The already published $1/d$ manuscript is included separately as `sources/prior-integrated-witness/paper.tex`, with its theorem and constants preserved. Its exact public revision and readable-source hashes are pinned in `PROVENANCE.json`. The received historical fallback differs only by disclosed editorial changes; its original hash is recorded separately. No inaccessible source is required for replay.

The pinned v3 source was read for the cone-law representation. The pinned quantitative supplement was read for normalization, first variation, and the every-maximum conversion. The actual pinned truncation `proof.tex` and exact checker were read. The new centering/brightness and projection-cap arguments above are self-contained quantitative estimates.

`check.exact.py` uses rational arithmetic and checks the barycentric weight correction, centered truncation cone laws, normalized and absolute projection identities, mixed-volume and projection scalings, both sets of explicit threshold constants, and finite near-permutation matrices. Its moment/witness regressions enumerate actual rational samples. The inherited truncation checker is replayed in a separate directory. Finite checks supplement the analytic proof and do not establish its arbitrary-body or nonatomic assertions. No inverse-Minkowski stability theorem, including the disputed proof of arXiv:2603.17726v3, is used.

References

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